Reg. No. | | | | | | | | | | |

[12ME73]

SEVENTH SEMESTER B.E EXAMINATION, DECEMBER-2015

CONTROL ENGINEERING

Time: 3 Hours **Maximum Marks:70**

Instructions:

Answer any **FIVE** full Questions.

1. a. Explain with example open and closed loop control system. **07**

b. Explain with a sketch proportional control. **07**

2. Obtain Transfer function of the mechanical system as shown in the fig. (2) **14**

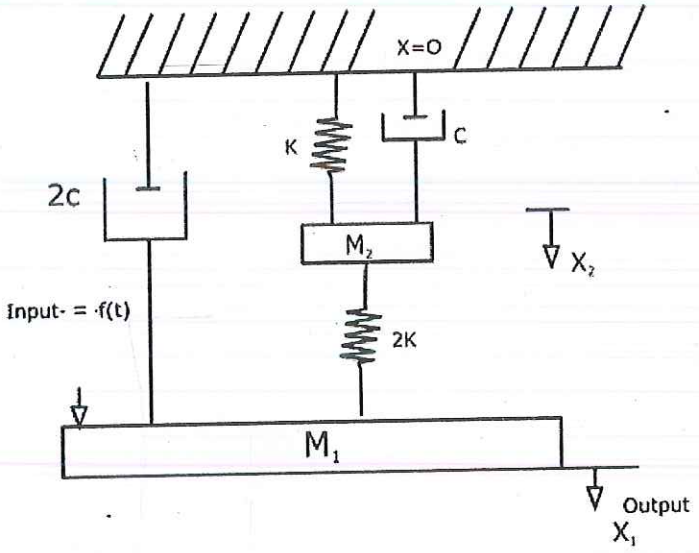

Fig-2

3. a. Using block diagram reduction technique. Find C/R as shown in Fig. 3(a) **08**

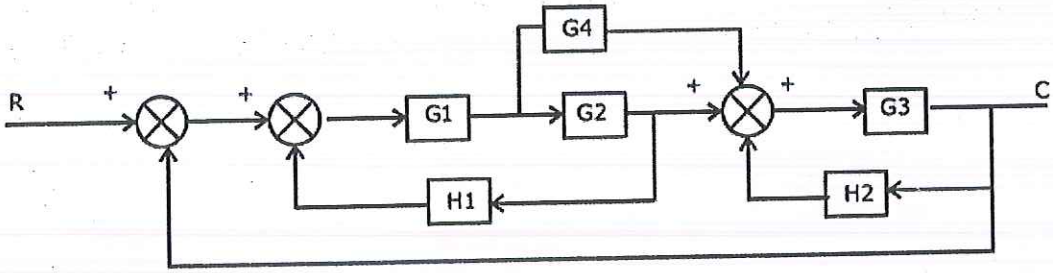

Fig-3a

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b. Explain Mason's gain formula used for solving signal flow graphs. **06**

4. Obtain the expression for rise time peak time, Peak over shoot and setting a second order system.

5. The open loop transfer function of a system is given by

$G(S)H(S) = \frac{K(S+2)}{S(S+6)(S+8)}$ sketch the root locus for the system.

6. Draw the complete Nyquist plot of the system whose OLTF is

$GH(S) = \frac{K}{(S+1)(S+2)(S+3)}$. Determine the range of K for which the system is stable.

7. Sketch the bode plot for unity feedback system given by

$G(S) = \frac{10(S+20)}{(S+1)(S+2)(S+3)}$. Determine Phase margin, gain margin, phase cross over frequency and gain crossover frequency.

8. Explain the following system of compensation. 1

i. Series compensation.

ii. Parallel compensation.